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RECYCLING OF LOW - DENSITY POLYETHYLENE (WATER SACHET) FILM AS FILLER FOR THE PRODUCTION OF POLYURETHANE IN COLD CHAIN SUPPLY MANAGEMENT.

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ABSTRACT

The aim of this study is to investigate the potentials of incorporating recycled low-density polypropylene (PE) sachet water film as a sustainable filler in the manufacture of flexible polyurethane (PU) foam materials that could enhance cold chain management systems. Different formulations of PU foam were tested, focusing on the effects of PE on key mechanical properties. All formulations were mixed for 15 seconds, ensuring a consistent process, though cream time (13.6-17.2 seconds) and rise time (108-124 seconds) varied across compositions. The foam densities ranged from 21.64 to 26.44 kg/m³, with higher PE content generally increasing density and Mechanical properties influenced by the PE content. Compression strength rose with higher PE but remained within the National Institute of Standards (NIS) limit of 10 kg/cm². However, flexural strength, tensile strength, force per unit area, and elongation decreased as PE content increased, indicating a trade-off between cost-efficiency and mechanical performance. Despite these changes, all formulations showed homogeneous morphology under Scanning Electron Microscopy, suggesting consistent material quality. The presented work indicates that PE can be efficiently introduced as a filler for flexible PU foams, hereby improving the foam performance, lowering production costs, and reducing the environmental impact of the indiscriminate disposal that is redirected by reusing these waste materials. However, higher PE percentages tend to lower the mechanical performance of the foam, indicating a need for optimized compositions.

Keywords: Polyurethane Foam, Low Density Polyethylene (LDPE), Water Sachet Film, Filler, Cold Chain.

1. INTRODUCTION

Certain additive materials have been used to impart foam plastics with cushioning and load bearing properties. Such additives include fillers either as bulk or reinforcing fillers. Fillers are substances added to other materials (plastics, composites and other materials) to lower consumption of the more expensive binder material or to improve some properties of the base material. In production of flexible polyurethane foams, fillers are used to modify the morphology and mechanical properties of the material. Predominantly, fillers are introduced to reduce the overall cost of the products, however the desired integrity in quality are sustained or improved upon as in case of extenders. (Dalen, 2014). Among over twenty most important fillers, calcite ($CaCO_3$) holds the largest market volume and is mainly used in plastic sectors. Other fillers include dolomite ($CaMg(CO_3)_2$), kaolin and talc.

Fillers are used in flexible urethane foams to reduce their flammability, increase the bulk and structural reinforcement. Fillers used in foam formulations are mainly inorganic in nature that needs vigorous agitation to produce a stable suspension. Inorganic solid materials used in filled foam formulations include; hydrated alumina, carbonates, silicates, silica, glass fibers, and barium sulfate. Historically, inorganic fillers such as barium sulfate and calcium carbonate have been used in flexible slab stock foams to achieve increased density and/or load bearing, and to reduce cost. Normal concentrations range from 20 to 150 parts per hundred parts polyols. The use of inorganic fillers has several disadvantages including difficulty of preparing and maintaining the dispersion; problems with removal of entrapped air; difficulty of mixing and pumping and extruding the filler/polyol slurry; the loss of the foam physical properties; difficulty of processing on all types of foam machinery, due to their abrasive nature, resulting in increased wear on machinery components.

Polyurethane foams (PURs) are generally obtained by poly-addition reaction between polyols, isocyanates and suitable blowing agents. Several additives (catalysts, surfactant and flame retardants) are added in order to tune the final characteristics of the product (Coccia, 2021). The mechanical and functional properties of the foams are strongly affected by the nature of polyols and isocyanate precursors as well as by the distribution of the soft and hard segments of PURs (Carriço, 2016; Coccia, 2021; Wendels, 2021).

Polyurethanes (PUR) are classified as polymeric materials, which are produced in a polyaddition reaction of multifunctional isocyanates with polyols. PUR materials may be obtained in the form of solid or foamed products with desired properties. Currently, PUR is produced on a large industrial scale and widely used in daily life due to their various and universal properties. (Ibrahim et al.; 2015, Aleksander et al., 2017) PUR most often are used in the furniture, automotive, footwear, construction, and refrigeration industries. There are multiple applications in which they appear in the form of foamed materials as flexible, viscoelastic, semi-rigid, and rigid foams. (Tejas, 2015; Prociak, 2016). Additionally, the local demand for cold chain products which includes essential food items (From the farm to the dining table) and drugs in the form of vaccines (From the research labs to the drug administration), it is firmly required that the items of interest must be stored at specified temperatures in order to ensure the shelf life of these very important commodities. This became more relevant especially in the advent of Covid -19 pandemic and the realities surrounding the demands for other vaccines in the fight against malaria, monkey pox, Anti-rabies, Lasa-fever and the likes as laid out by the Centre for Disease Control vaccine storage and Handling toolkit, March 2024. The integrity of the aforementioned products is linked to a sustained cold chain delivery as amplified in World Health Organization Unit 4: Cold Chain and logistics management, which has a large bearing on the paddings and laggings derived from desired polyurethanes products. (polyurethanes.org). The recycling of polyethylene films from pure water sachet water as a desirable filler in production of polyurethanes is a major step in the forward direction in resolving the impact of indiscriminately disposed LDPE pure water sachet film. The menace cursed by these littering films includes the blockage of drainages and water ways that results in enhancing destructive floods and the development of stagnant ponds that serve as habitat for breeding undesirable organisms that are vectors to malaria, cholera and other parasites. Our environment and our health are positively impacted by the innovation of this research as supported by Singh et al., 2017. Any Patent from the outcome of the findings herein is a viable economic venture. One of the several targets for the product engineered is in the manufacture of enhanced customized units for cold chain management supply system. In recent times, the movement of items such as food, drugs, especially vaccines that require storage at low temperatures in order to retain product integrity has become increasingly unavoidable globally. To this end the World Health Organization in collaboration with Pan American Health Organization makes every effort to ensure that supplies such as drugs and vaccines supplies, provide the expected benefits when an individual gets immunized, it is required that epidemiologist and health staff are equipped to provide potent supply for protecting the recipient against the targeted disease(s). In this respect, the focus of the Immunization program should be built around a minimum of five pillars: (1) Countries are advised to use good quality refrigeration equipment (2) Health staff involved in managing vaccines should be trained (3) Cold Chain and supply chain operations should be evaluated (4) Research and development should be carried out in the area of cold chain technology with respect to soft and hard wares (5) The management capacity and skills to support all operations linked to cold chain and supply chain needs to be improved to accommodate new products. (PAHO & WHO, Cold Chain 2025)

2. MATERIALS AND METHODS

2.1 SAMPLE COLLECTION

Industrial chemicals such as: polymer polyol (PPO), conventional polyol (CPO), toluene di-isocyanate (TDI) (80/20), stannous octanoate, dimethyl ethanolamine, polysilicone methylene dichloride, and $CaCO_3$ were products of Korean Fire Chemicals Ltd., BASF and Dupont, and were graciously provided by Vitafoam Nig. Plc, Jos Mega Factory. Sachet water film was collected, washed and dried, and was pulverized at Destiny Recycling Company Kwan Jos Plateau State.

2.2 SAMPLE TREATMENTS

Low density polyethylene (LDPE) based water sachet films were recycled into polymer powder by thermochemical treatment. Varying weights of sachet films were dissolved in automotive gas oil (AGO) at elevated temperature, followed by quenching in an ice-bath stainless steel container to determine optimum powder yield and particle size distribution. Sieve analysis of powder was carried out into different particle sizes and consequently to eliminate lumps**.**

2.3 FOAM FORMULATION

One-shot technique of urethane foam formulation was used (Richter, 1974). The ingredients were measured accurately using syringes, micro-syringes, pipettes and measuring cylinders into moulds of dimensions (21.5cm x 13.7cm x 14.5cm), using the recipes on Table 1. The components were thoroughly stirred and TDI added with continuous stirring until the systems creamed. The foams were then allowed to rise undisturbed and left to cure 24 hours after which physico- mechanical properties were determined.

Table 1: **Formulation's ratio of PUFs**

Sample Identity	0%	5.0%	7.5%	10%	12.5%	15%
CPO (g)	200	200	200	200	200	200
$CaCO_3$ or PE (g)	0	10	15	20	25	30
Silicone (g)	1.80	1.80	1.80	1.80	1.80	1.80
Amine (g)	0.40	0.40	0.40	0.40	0.40	0.40
Stannous (g)	0.38	0.38	0.38	0.38	0.38	0.38
Water (g)	8.64	8.64	8.64	8.64	8.64	8.64
TDI (g)	110.96	110.96	110.96	110.96	110.96	110.96

2.4 PHYSICAL PROPERTIES

Some of the polyurethane foam's physical properties were found in a rather simple fashion. Densities for each type of foam were found using a scale and a set of calipers in accordance with ASTM D3574-95, "Standard Test Methods for Flexible Cellular Materials – Slab, Bonded, and Molded Urethane Foams," Test A. There were three samples weighed and measured for each type of foam and an average density was determined. The density for each type of polyurethane foam tested is listed in Table I.

2.5 MICROSTRUCTURE DETERMINATION

The samples were cut gently into thin wafers using a fine blade. These wafers were later mounted on a stub and coated with silver paint at the edges. The silver paint is used to provide a conductive path from the surface of the coated sample to the stub, which then goes off to an electrical ground. Later the sample was placed in an oven at 150°F for 5–20 min for drying the paint. Since the samples are non-conductive, either sputter or carbon coating was employed. Then the samples were put under the scanning electron microscope for analysis. Two scanning electron microscopes were used: an AMRAY 1600 with an accelerating voltage of 20 kV and a Hitachi S-4700 FESEM with an accelerating voltage of 5 kV.

2.6 TENSION TEST

The purpose of conducting a tension test was to determine the tensile strength of each type of polyurethane foam. This test also gives an elastic modulus for the foam in tension. The procedure for tension testing was in accordance with ASTM D3574-95, "Standard Test Methods for Flexible Cellular Materials – Slab, Bonded, and Molded Urethane Foams," Test E. Test specimen dimensions are shown in the ASTM standard. A special large extension clip gage of 25-mm gage length was used in the middle of the specimen to measure the elongation during the tension tests.

2.7 COMPRESSION TEST

The purpose of conducting a compression test was to determine a compressive stress-strain relationship for the different types of reinforced polyurethane foam. Three samples of each type of foam were tested to give an average and to show the repeatability of data. This test also gives the compressive elastic modulus. The procedure for the compression test was in accordance with ASTM D3574-95, "Standard Test Methods for Flexible Cellular Materials – Slabs, Bonded, and Molded Urethane Foams," Test C. Pieces were cut from a polyurethane foam plate with a width and a length of 20.0-mm (0.787-inch). Two of the pieces were placed together to make a 25.4-mm (1.0 inch) thick specimen, and then tested as directed by the standard. There were three specimens tested. Each specimen was preflexed twice by compressing it 75 to 80% of its original thickness, in this case about 5 mm. The specimen was then relaxed for 6 min. Then the specimen was compressed to about 25% of its thickness at a rate of 0.83 mm/s (0.033 in/s)

3 RESULTS AND DISCUSSION

RESULTS

Table 2: Density of various reinforced polyurethane foam

PUFs	Mixed Time (s)	Cream Time (s)	Rise Time (s)	Density (kg/m^3)
0% $CaCO_3$	15	14.8	111	21.64
5% $CaCO_3$	15	13.6	120	25.76
5% PE	15	15.4	108	24.28
10% PE	15	16.5	124	25.27
15% PE	15	16.8	114	25.36
20%	15	17.2	109	26.44

Table 3: Mechanical properties of various reinforced polyurethane foam.

PUFs	**Compression (kg/cm^2)**	**NIS Value for Compression (kg/cm^2) (Max)**	**Flexural Strenght (kg/cm^2)**	**Tensil Strain (N/mm^2)**	**Tensil Stress (N/mm^2)**	**Elongation (N/mm^2)**
0% $CaCO_3$	9.0	10	41.7	2.431	33.806	3.646
5% $CaCO_3$	10.7	10	37.9	1.595	22.512	2.306
5% PE	10.9	10	34.9	1.50	21.50	2.05.
10% PE	11.9	10	28.4	1.46	20.37	1.98
15% PE	12.0	10	12.18	1.37	18.76	1.55
20% PE	12.0	10	12.1	0.9	16.59	0.60

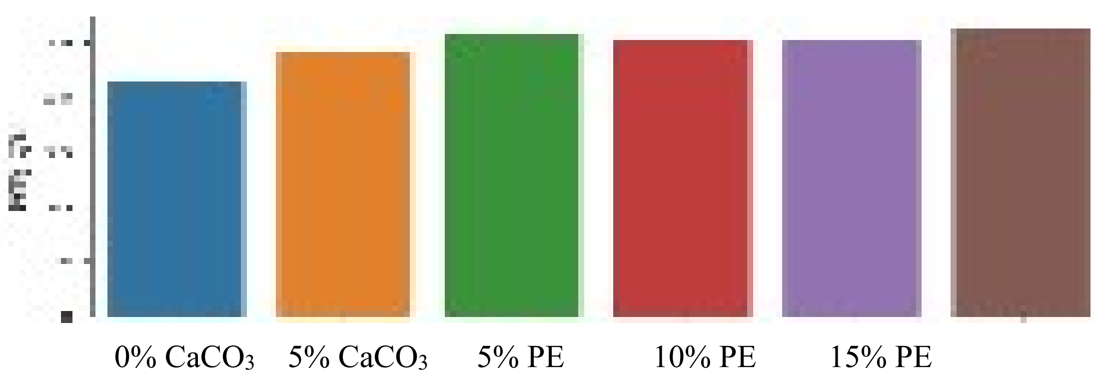
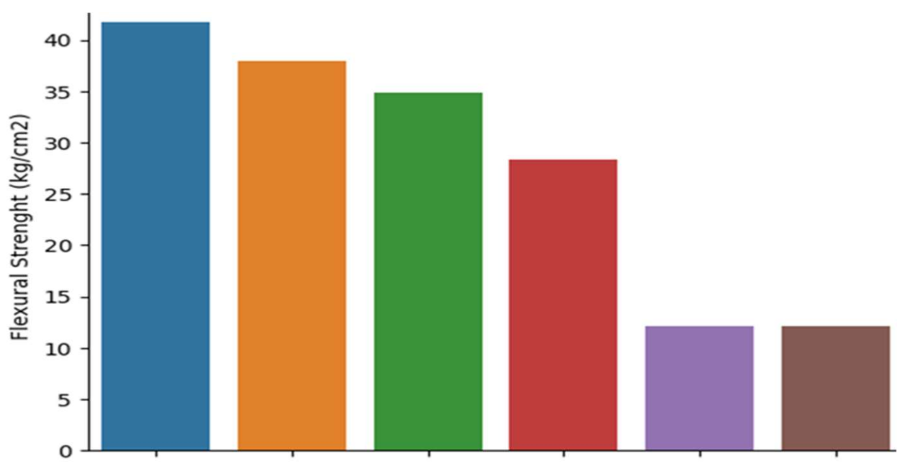


Figure 1: Density of PUFs

0% $CaCO_3$ 5% $CaCO_3$ 5% PE 10% PE 15% PE
20% PE

Figure 2: Cream Time of PUFs



0% $CaCO_3$ 5% $CaCO_3$ 5% PE 10% PE 15% PE 20% PE

Figure 3: Flexural Strength of PUFs

0% $CaCO_3$ 5% $CaCO_3$ 5% PE 10% PE 15% PE 20% PE

Figure 4: Compression of PUFs

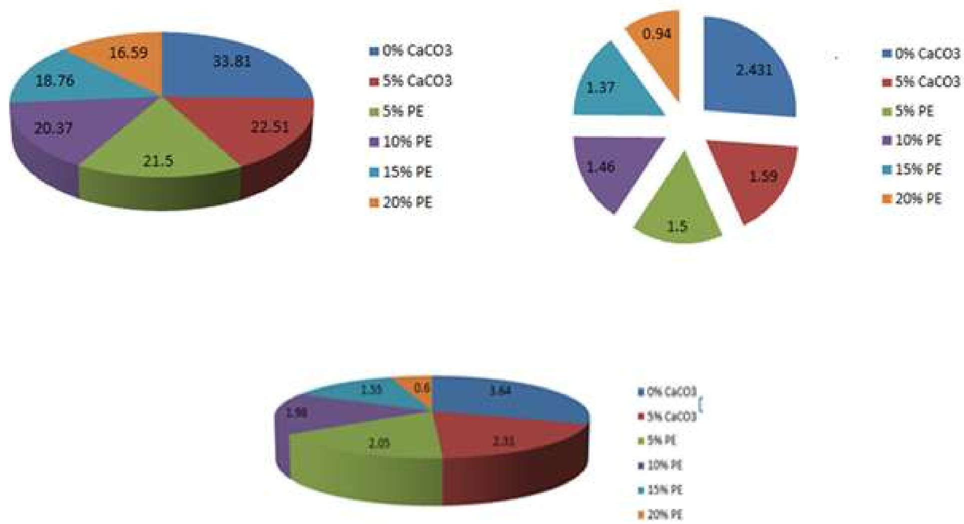


Figure 5: (a) Tensile Stress of PUFS, (b) Tensile Strain of PUFs and (c) Elongation of PUFs

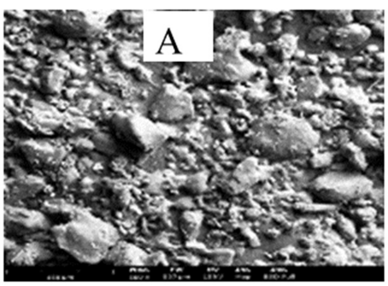


Plate 1: A – 0% Filler

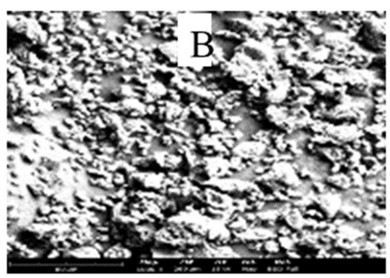


Plate 2: B – 5% $CaCO_3$

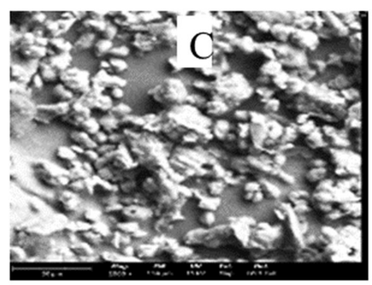


Plate 3: C – 5% PE

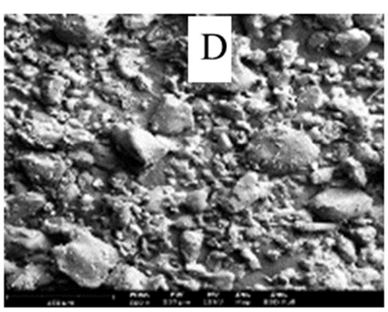


Plate 4: D - 10% PE

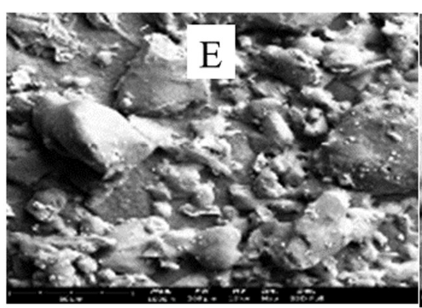


Plate 5: E - 15% PE

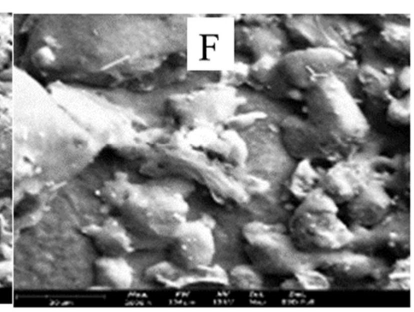


Plate 6: F - 20% PE

3.1 DISCUSSION

Table 1 provides data on various polyurethane foams (PUFs) with different compositions, specifically varying percentages of calcium carbonate ($CaCO_3$) and polyethylene (PE).

Various forms of polyurethane foam, describing their ingredients. The duration for blending the foam components. In all instances, it remains consistent at 15 seconds, implying a uniform mixing time for the foams. The time it takes for the foam to reach a creamy consistency after mixing. It varies for different formulations, with values ranging from 13.6 to 17.2 seconds. The time it takes for the foam to rise after reaching the creamy consistency. It also varies across different formulations, with values ranging from 108 to 124 seconds. The density of the formed polyurethane foam, measured in kilograms per cubic meter (kg/m^3). The density varies across different formulations, with values ranging from 21.64 to 26.44 kg/m^3. The filler did not show any significant effect on density of the flexible foams. This relates to a study by Javni (2002), in which micro – silica filler was used.

Table 2 shows how changing the composition of the polyurethane foam affects its cream time, rise time, and final density. For example, increasing the percentage of polyethylene (PE) generally results in a slightly higher density, and there are variations in the cream time and rise time as well. The consistent mixed time suggests a standardized mixing process across all formulations.

The kg/cm^2 measurement of the polyurethane foam determines its compression strength, indicating the highest level of perplexity and burstiness. It represents the maximum load the material can withstand under compressive forces. As the percentage of PE increases, the compression strength tends to increase. The maximum value allowed for compression strength according to the National Institute of Standards (NIS). In this case, the value is consistently 10 kg/cm2 for all formulations. Latinwo et al., (2010) suggested that the increase in the hardness of the foam by nano and micro fillers is due mainly to the random dispersion of the fillers into the polyurethane matrix. The interaction between the filler particles and the polymer matrix as a result of intercalation of the polymer chains into the filler galleries influence the chemical structures of the polymer chains. The strength of the material when subjected to bending or flexural forces. It decreases with an increase in the percentage of PE in the formulation. The ability of the material to withstand stretching or tensile forces. It generally decreases with an increase in the percentage of PE. This relate to a study by Zukowska et al., (2022), which ground tire rubber (GTR) was used as filler in flexible polyurethane foams, which shows it can be efficiently applied as a filler for flexible PU foams, which could simultaneously enhance their performance. The force per unit area applied to the material during tensile testing. It shows a decreasing trend as the percentage of PE increases. The measure of how much a material can stretch or deform before breaking. It decreases with an increase in the percentage of PE. the table indicates how changing the composition of the polyurethane foam affects its compression strength, flexural strength, tensile properties, and elongation. Generally, higher percentages of polyethylene (PE) result in lower mechanical properties for these specific foam formulations. They all have homogenous morphology.

4 CONCLUSION AND RECOMMENDATIONS

4.1 CONCLUSION

The use of water sachet waste film powder as filler in polyurethane foam production has proven to be an efficient in the production of polyurethane foams and it presents an environmentally friendly way of disposal of this waste, and it will go a long way in addressing the menace generated by the littering of these waste films. The soil fertility degradation as a result of the poor aeration and poor water penetration caused by these materials will be redirected thereby, improving compost for agriculture by separation of the waste films from other municipal solid wastes which are biodegradable.

Notably, the recycling LDPE into powder for the production of flexible polyurethane Foam can contribute to resource conservation. It allows for the reuse of plastic material, reducing the demand for new raw materials. Also, the services of people for the collection, sorting and cleaning of the wastes will be needed and as such, job opportunities will be created

It is worthy to state that the material generated is readily explorable for the production of insulators and packaging products in cold chain boxes and appliances that will enhance cold chain supply.

4.2 RECOMMENDATION

Further study should be carried out in combining *LDPE sachet water* with other types of polymers to investigate if it could provide a more effective and improved foam formulations suitable for conformations and templates that would present avenue for cold chain supply

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